HW 9

1. $A \in \mathbb{R}^{3\times 2}$ $\quad A = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \end{pmatrix}$

$A = uv^T$

$r = \text{rank}(A) = 1$ since $A = uv^T$, u and v nonzero

$\sigma_1 = \|A\| = \|uv^T\| = \|u\| \cdot \|v\| = \sqrt{(1+4+1)(1+1)} = \sqrt{12} = 2\sqrt{3}$

$u_1 = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \quad v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$$A = U\Sigma V^T = \boxed{\begin{bmatrix} 1/\sqrt{6} \\ 2/\sqrt{6} \\ -1/\sqrt{6} \end{bmatrix} \begin{bmatrix} 2\sqrt{3} \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}}$$

2. Let A^+ be the pseudoinverse of $A \in \mathbb{R}^{m \times n}$.

a) $m \geq n$, A full rank $\qquad (\Sigma^+\Sigma)_{ii} = \frac{1}{\sigma_i} \cdot \sigma_i = 1$

$$A^+A = V\Sigma^+U^TU\Sigma V^T = V\begin{bmatrix} 1/\sigma_1 & & & & \\ & 1/\sigma_2 & & & \\ & & \ddots & & \\ & & & 1/\sigma_r & \\ & & & & 0 \end{bmatrix}\begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_r \end{bmatrix}V^T = VIV^T = I$$

$$AA^+ = U\Sigma V^TV\Sigma^TU^T = U\begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_n \end{bmatrix}\begin{bmatrix} 1/\sigma_1 & & & & \\ & \ddots & & & \\ & & 1/\sigma_r & & \\ & & & 0 & \\ & & & & \ddots \; 0 \end{bmatrix}U^T = UIU^T = I$$

b) $(A^+)^+ = A$

From part (a), we get $A^+A = I \Rightarrow A^+ = A^{-1}$. Thus,

$$(A^+)^+ = (A^{-1})^+ = (A^{-1})^{-1} = A \quad \checkmark$$

Where the pseudoinverse is the left inverse of A

3. Let $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_k \geq 0$ be the singular values of $A \in \mathbb{R}^{m \times n}$, $k = \min(m,n)$.

Show $\|A\|_F = \sqrt{\sum_{i=1}^{k} \sigma_i^2}$

a) $\|A\|_F = \sqrt{\operatorname{trace}(A^T A)}$ using $\|A\|_F = \sqrt{\sum_{i,j} a_{ij}^2}$

$$(A^T A)_{j\ell} = \sum_{i=1}^{n} a_{ij} \cdot a_{i\ell}$$

goes by row

$$\operatorname{trace}(A^T A) = \sum_{j=1}^{n} (A^T A)_{jj} = \sum_{j=1}^{n} \sum_{i=1}^{n} a_{ij}^2 = \sum_{i,j} a_{ij}^2$$

$$\|A\|_F = \sqrt{\operatorname{trace}(A^T A)} = \sqrt{\sum_{i,j} a_{ij}^2}$$

b) $\operatorname{trace}(A^T A) = \operatorname{trace}(V \Sigma^T \Sigma V^T) = \operatorname{trace}(\Sigma^T \Sigma V V^T) = \operatorname{trace}(\Sigma^T \Sigma) = \sum_{j=1}^{k} \sigma_j^2$ ✓

V diagonalizes $A^T A$

4. Find optimal rank 2 approx. A_2 to matrix $A = \begin{pmatrix} 1 & 6 & 2 \\ -4 & 0 & 4 \\ 5 & 0 & 4 \end{pmatrix}$

Eckart-Young Thm, in spectral norm,

$$A_2 = \underline{\sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T}$$

$$\|A - A_2\|_2 = \sigma_3$$

Use AA^T to compute singular values of A.

$$AA^T = \begin{pmatrix} 1 & 6 & 2 \\ -4 & 0 & 4 \\ 5 & 0 & 4 \end{pmatrix} \begin{pmatrix} 1 & -4 & 5 \\ 6 & 0 & 0 \\ 2 & 4 & 4 \end{pmatrix} = \begin{pmatrix} 41 & 0 & 21 \\ 0 & 32 & 0 \\ 21 & 0 & 41 \end{pmatrix}$$

$$\det(AA^T - \lambda I) = \begin{vmatrix} 41-\lambda & 0 & 21 \\ 0 & 32-\lambda & 0 \\ 21 & 0 & 41-\lambda \end{vmatrix} \Rightarrow \lambda_1 = 32 \Rightarrow \begin{vmatrix} 41-\lambda & 21 \\ 21 & 41-\lambda \end{vmatrix} \Rightarrow \lambda = 41 \pm 21 \Rightarrow \lambda = 62, 20$$

$$\Rightarrow \sigma_1 = \sqrt{62}, \quad \sigma_2 = \sqrt{32}, \quad \sigma_3 = \sqrt{20}$$

$$\therefore \|A - A_2\|_2 = \sqrt{20} = \boxed{2\sqrt{5}}$$

5. a) Let $A = QQ^TA$, $B = Q^TA$

SVD of B: $B = \hat{U}_k \Sigma_k V_k^T$

Set $U_k = Q\hat{U}_k$

$$U_k \Sigma_k V_k^T = Q\hat{U}_k \Sigma_k V_k^T = QB = Q(Q^TA) = A$$

U_k has orthonormal columns since Q and $\hat{U}_k$ both have orthonormal columns and $Q\hat{U}_k$ preserves orthonormality, resulting in the exact SVD output.

b) Flop Count

step	cost
$A*\Omega$	$O(mnk)$
qr(Y,'econ')	$O(mk^2)$
Q'*A	$O(mnk)$
svd(B,'econ')	$O(nk^2)$
$U_k = Q\hat{U}_k$	$O(mk^2)$

Total $= O(mnk + nk^2)$

This is more efficient and cheaper than $O(mn^2)$ because when $k < n$, the number of flops decreases by a factor of roughly k/n.

c) We should choose k by capturing where the optimal error drops below the desired tolerance.

- by $k \approx 10$, errors are already $\sim 10^{-7}$
- by $k \approx 15-20$, close to ε_{mch}

Since randomized error is within a small constant around the optimal, we should choose k such that σ_{k+1} is below the tolerance.

```python
import numpy as np
from scipy.linalg import hilbert
import matplotlib.pyplot as plt


def randomized_svd(A, k):
    """
    Randomized SVD algorithm.
    Returns U_k, Sigma_k, V_k such that A ≈ U_k @ diag(Sigma_k) @ V_k.T
    """
    m, n = A.shape
    # Step 1: Random sketch
    Omega = np.random.randn(n, k)
    Y = A @ Omega                        # m x k

    # Step 2: QR of Y (economy)
    Q, _ = np.linalg.qr(Y)              # Q is m x k

    # Step 3: Project A
    B = Q.T @ A                          # k x n

    # Step 4: SVD of small matrix B
    U_hat, Sigma_k, Vt_k = np.linalg.svd(B, full_matrices=False)

    # Step 5: Recover left singular vectors
    U_k = Q @ U_hat                      # m x k

    return U_k, Sigma_k, Vt_k


def spectral_norm_error(A, U_k, Sigma_k, Vt_k):
    """||A - U Sigma V^T||_2 = largest singular value of the residual"""
    A_approx = (U_k * Sigma_k) @ Vt_k
    residual = A - A_approx
    return np.linalg.norm(residual, ord=2)


# ── Build 100x100 Hilbert matrix ──────────────────────────────────────
n = 100
H = hilbert(n)

# ── True singular values (for Eckart-Young optimal errors) ────────────
_, true_sigmas, _ = np.linalg.svd(H)

k_values = [1, 2, 3, 5, 8, 10, 15, 20]
rand_errors = []
optimal_errors = []

np.random.seed(42)
for k in k_values:
    # Randomized SVD error
    U_k, Sigma_k, Vt_k = randomized_svd(H, k)
    rand_errors.append(spectral_norm_error(H, U_k, Sigma_k, Vt_k))

    # Eckart-Young optimal error = sigma_{k+1}
    optimal_errors.append(true_sigmas[k] if k < n else 0.0)

print(f"{'k':>4} | {'Randomized Error':>18} | {'Optimal (σ_{k+1})':>18} |
{'Ratio':>8}")
print("-" * 58)
for k, re, oe in zip(k_values, rand_errors, optimal_errors):
    ratio = re / oe if oe > 1e-16 else float('inf')
    print(f"{k:>4} | {re:>18.6e} | {oe:>18.6e} | {ratio:>8.3f}")

"""OUTPUT
   k |   Randomized Error |  Optimal (σ_{k+1}) |    Ratio
----------------------------------------------------------
   1 |       2.142902e+00 |       8.214456e-01 |    2.609
   2 |       2.420872e-01 |       2.185959e-01 |    1.107
   3 |       9.509982e-02 |       4.929225e-02 |    1.929
   5 |       3.233078e-02 |       1.885063e-03 |   17.151
   8 |       4.322691e-05 |       8.536281e-06 |    5.064
  10 |       7.015601e-07 |       1.788722e-07 |    3.922
  15 |       3.334944e-11 |       5.212225e-12 |    6.398
  20 |       2.489985e-15 |       1.555226e-16 |   16.010
"""
```